

EMDEN, Germany

WMO: 102000

Lat: 53.3881N

Lon: 7.2286E

Elev: -5

StdP: 14.70

Time Zone: 1.00 (EUC)

Period: 97-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.4	21.9	13.7	11.0	20.0	17.6	13.4	23.4	29.6	44.1	26.3	44.6	7.4	90	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	81.7	67.4	77.6	66.1	74.0	64.3	69.1	78.3	67.3	75.1	65.6	72.0	9.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	95.6	72.1	64.4	90.4	71.0	62.9	85.8	68.8	33.2	78.4	31.7	75.4	30.4	72.2	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.4	21.4	19.0	DB	12.1	89.0	6.4	2.6	7.5	90.8	3.7	92.4	0.1	93.8	-4.5	95.7
			WB	11.7	73.3	6.4	1.3	7.1	74.2	3.3	74.9	-0.3	75.6	-5.0	76.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.4	36.9	37.1	41.1	47.9	54.2	59.2	63.2	63.3	58.4	50.5	43.2	37.4
	DBStd	11.49	7.63	7.23	6.36	5.94	5.41	5.32	4.83	4.56	4.73	6.11	6.16	7.41
	HDD50	1853	406	364	280	108	20	1	0	0	2	69	211	392
	HDD65	5778	870	782	741	512	337	187	96	89	206	449	653	856
	CDD50	1648	1	1	4	46	150	278	409	411	253	85	9	1
	CDD65	100	0	0	0	0	3	15	40	35	7	0	0	0
	CDH74	791	0	0	0	1	34	141	323	246	46	0	0	0
	CDH80	177	0	0	0	0	1	32	77	64	3	0	0	0
(14) Wind	WSAvg	9.3	10.7	10.6	10.4	9.1	8.8	8.5	8.3	7.9	8.2	9.6	9.2	9.9
(15) Precipitation	PrecAvg	30.9	2.5	1.7	2.2	1.8	2.2	2.8	3.1	2.9	2.9	2.6	3.3	2.7
	PrecMax	37.2	5.5	4.5	5.0	4.1	5.9	5.7	6.7	6.8	7.7	5.2	5.4	6.3
	PrecMin	23.5	0.1	0.2	0.4	0.3	0.4	1.1	0.4	0.5	0.6	0.6	0.8	0.6
	PrecStd	3.7	1.3	0.9	1.1	1.0	1.1	1.3	1.6	1.5	1.5	1.2	1.1	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	54.7	61.0	71.9	78.5	84.8	86.3	86.4	79.3	67.7	59.4	54.1
		MCWB	51.0	50.8	52.0	57.7	63.6	69.6	69.0	68.5	66.7	61.9	55.5	52.2
	2%	DB	51.0	51.6	56.8	67.6	73.3	78.1	81.9	80.4	74.2	63.6	55.6	51.3
		MCWB	49.0	48.7	50.5	55.8	60.7	65.3	67.4	68.0	64.8	59.2	53.2	49.5
	5%	DB	48.5	49.1	53.4	63.0	68.9	73.0	77.4	75.5	69.9	60.9	53.4	49.1
		MCWB	46.7	47.0	49.1	54.3	58.3	63.5	66.5	66.2	62.9	57.5	51.1	47.4
	10%	DB	46.7	46.8	50.6	59.1	64.9	69.0	73.3	71.8	66.2	58.9	51.3	47.1
		MCWB	45.1	44.8	47.2	52.4	56.8	61.5	64.8	64.1	60.7	56.0	49.2	45.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.4	51.8	54.6	60.0	66.3	70.6	71.6	71.5	68.6	62.4	56.2	52.8
		MCDB	53.0	54.3	58.9	68.1	75.3	81.3	80.8	81.3	75.0	66.9	58.4	53.7
	2%	WB	49.3	49.3	51.6	57.1	62.4	67.2	69.1	68.8	65.9	59.8	53.5	49.8
		MCDB	50.9	50.8	54.5	64.3	70.3	75.4	78.8	77.1	72.8	62.6	55.4	51.0
	5%	WB	47.1	47.1	49.7	55.1	59.7	64.6	67.3	66.6	63.5	58.1	51.2	47.5
		MCDB	48.4	48.9	53.3	61.5	66.2	70.5	75.6	73.3	68.8	60.4	52.9	48.9
	10%	WB	45.1	45.0	47.3	52.9	57.5	62.4	65.3	65.0	61.5	56.4	49.5	45.6
		MCDB	46.5	46.5	50.2	58.6	63.6	68.0	71.8	70.6	65.3	58.6	51.0	46.9
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.6	11.6	15.6	15.6	14.7	14.0	13.8	13.3	11.2	8.1	7.4
		MCDBR	7.8	9.7	17.0	23.6	23.9	22.8	23.4	21.2	19.9	12.7	9.0	7.7
		MCWBR	7.3	7.9	11.8	13.9	13.2	12.0	11.1	10.7	11.5	9.1	7.6	7.5
	5% WB	MCDBR	7.7	8.4	15.6	20.8	20.5	19.3	20.8	18.5	18.4	12.4	8.8	7.6
		MCWBR	7.4	7.3	11.4	13.1	12.7	11.4	10.8	10.1	11.3	9.5	7.8	7.6
(40) Clear-Sky Solar Irradiance	taub		0.328	0.347	0.375	0.407	0.404	0.403	0.415	0.410	0.388	0.362	0.342	0.320
	taud		2.317	2.331	2.272	2.237	2.284	2.311	2.290	2.318	2.373	2.404	2.350	2.309
	Ebn at Noon		201	236	253	260	268	269	263	257	248	228	193	179
	Edh at Noon		19	26	33	39	39	39	39	36	30	24	19	16
(44) All-Sky Solar Radiation	RadAvg		186	392	780	1300	1614	1696	1607	1335	947	529	227	142
	RadStd		18	61	84	127	134	123	187	108	95	54	22	16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-220	N/A	N/A

Nomenclature: See separate page